

CO3-AT1-DS-CASE STUDY

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Case Study 1 — Employee Salaries

Salaries (₹ Lakhs):

4, 5, 5, 6, 7, 8, 8, 9, 30

Number of observations:

1. Mean, Median and Mode

(A) Mean

Formula

Answer:

Mean = ₹9.11 Lakhs

(B) Median

Since (Odd),

Median Position

5th Observation = 7

Answer:

Median = ₹7 Lakhs

(C) Mode

Salary	Frequency
4	1
5	2
6	1
7	1
8	2
9	1
30	1

Answer:

Mode = ₹5 Lakhs and ₹8 Lakhs (Bimodal)

2. Range

Formula

Answer:

Range = ₹26 Lakhs

3. Variance and Standard Deviation

Mean = 9.11

X	X-9.11	(X-9.11) ²
4	-5.11	26.12
5	-4.11	16.90
5	-4.11	16.90
6	-3.11	9.68
7	-2.11	4.46
8	-1.11	1.23
8	-1.11	1.23
9	-0.11	0.01
30	20.89	436.23

Variance

Variance = 56.97

Standard Deviation

Standard Deviation = 7.55 Lakhs

4. Outlier

Using observation,

Outlier = ₹30 Lakhs

(It is much higher than the remaining salaries.)

5. Best Measure of Central Tendency

Since the dataset contains an **outlier (₹30 Lakhs)**, the **Median (₹7 Lakhs)** best represents the typical salary because it is not affected by extreme values.

Conclusion

- Mean = ₹9.11 Lakhs
- Median = ₹7 Lakhs
- Mode = ₹5 Lakhs & ₹8 Lakhs
- Range = ₹26 Lakhs
- Variance = 56.97
- Standard Deviation = 7.55 Lakhs
- **Median** is the best measure due to the outlier.

Case Study 2 — Website Response Time Response Times

120, 125, 128, 130, 132, 135, 140, 145, 150, 300

Number of observations:

1. Mean and Median

(A) Mean

Formula

Answer:

Mean = 150.5 ms

(B) Median

Since (Even),

Middle values = 5th and 6th observations

Answer:

Median = 133.5 ms

2. Range

Formula

Answer:

Range = 180 ms

3. Variance and Standard Deviation

Mean = 150.5

X	$X - 150.5$	$(X - 150.5)^2$
120	-30.5	930.25
125	-25.5	650.25
128	-22.5	506.25
130	-20.5	420.25
132	-18.5	342.25
135	-15.5	240.25
140	-10.5	110.25
145	-5.5	30.25
150	-0.5	0.25
300	149.5	22350.25

Variance

Variance = 2558.15

Standard Deviation

Standard Deviation = 50.58 ms

4. Skewness

Since

- Mean = 150.5
- Median = 133.5

Answer:

The data is positively (right) skewed due to the high response time of **300 ms**.

5. Why is Median More Useful?

The value 300 ms is an outlier that increases the mean.

The median (133.5 ms) is not affected by the outlier and better represents the typical website response time.

Conclusion

- **Mean = 150.5 ms**
- **Median = 133.5 ms**
- **Range = 180 ms**
- **Variance = 2558.15**
- **Standard Deviation = 50.58 ms**
- The distribution is right-skewed, and the median is the better measure of central tendency.

Case Study 3 — Student Performance

Marks:

42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

Number of observations:

1. Find Q_1 , Q_2 and Q_3

Q_2 (Median)

Since (Even),

Middle values = 6th and 7th observations

Q_1

Lower half:

42, 45, 48, 50, 52, 52

Q_3

Upper half:

55, 58, 60, 62, 65, 95

Answer:

- **$Q_1 = 49$**
- **$Q_2 = 53.5$**
- **$Q_3 = 61$**

2. Calculate the IQR

Formula

Answer:

IQR = 12

3. Identify Outliers ($1.5 \times \text{IQR}$ Rule)

Lower Limit

Upper Limit

Acceptable Range:

Since

Answer:

Outlier = 95

4. Mean and Median

Mean

Mean = 57

Median

Median = 53.5

5. Effect of the Outlier

The value 95 is an outlier that increases the mean from the central values, while the median (**53.5**) remains almost unaffected.

Therefore, the median better represents the typical student performance.

Conclusion

- **$Q_1 = 49$**
- **Q_2 (Median) = 53.5**
- **$Q_3 = 61$**
- **IQR = 12**
- **Outlier = 95**
- **Mean = 57**
- **Median = 53.5**
- The outlier 95 increases the mean, so the median is a better measure of central tendency.

Case Study 4 — E-Commerce Orders

Orders received over 10 days:

18, 20, 22, 22, 24, 25, 26, 28, 30, 35

Number of observations:

1. Calculate Mean, Median and Mode

(A) Mean

Formula

Answer:

Mean = 25 Orders

(B) Median

Since (Even),

Middle values = 5th and 6th observations

Answer:

Median = 24.5 Orders

(C) Mode

Orders	Frequency
18	1
20	1
22	2
24	1
25	1
26	1
28	1
30	1
35	1

Answer:

Mode = 22 Orders

2. Variance and Standard Deviation

Mean = 25

X	X-25	(X-25) ²
18	-7	49
20	-5	25
22	-3	9
22	-3	9
24	-1	1
25	0	0
26	1	1
28	3	9
30	5	25
35	10	100

Variance

Variance = 22.8

Standard Deviation

Standard Deviation = 4.77 Orders

3. Coefficient of Variation (CV)

Formula

Answer:

Coefficient of Variation = 19.08%

4. What does the Standard Deviation tell the Company?

The standard deviation of **4.77** orders indicates that the daily orders vary by about 5 orders from the average. This helps the company understand the consistency of daily order volume.

5. Is the Data Highly or Moderately Variable?

Since the **CV = 19.08%** (less than 20%), the order data shows moderate variability and is fairly consistent.

Conclusion

- **Mean = 25 Orders**
- **Median = 24.5 Orders**
- **Mode = 22 Orders**
- **Variance = 22.8**
- **Standard Deviation = 4.77 Orders**
- **Coefficient of Variation = 19.08%**
- The order data has moderate variability.

Case Study 5 — Comparing Two Data Science Models

Model A: 2, 3, 3, 4, 4, 5, 5, 6

Model B: 1, 1, 2, 4, 5, 7, 8, 9

Number of observations:

1. Calculate the Mean Error

Mean Error (Model A) = 4

Mean Error (Model B) = 4.625

2. Calculate the Variance

Model A, Mean = 4

X	X-4	(X-4) ²
2	-2	4
3	-1	1
3	-1	1
4	0	0
4	0	0
5	1	1
5	1	1
6	2	4

Variance (Model A) = 1.5

Model B

Mean = 4.625

X	$X-4.625$	$(X-4.625)^2$
1	-3.625	13.14
1	-3.625	13.14
2	-2.625	6.89
4	-0.625	0.39
5	0.375	0.14
7	2.375	5.64
8	3.375	11.39
9	4.375	19.14

Variance (Model B) = 8.73

3. Calculate the Standard Deviation

Model A

Standard Deviation (Model A) = 1.22

Model B

Standard Deviation (Model B) = 2.95

4. Which Model Has More Consistent Predictions?

Since

- **Variance (A) = 1.5 < 8.73**
- **SD (A) = 1.22 < 2.95**

Answer:

Model A has more consistent predictions because it has a lower variance and standard deviation.

5. Can the Model with the Lower Mean Error Be Considered Better?

No.

A lower mean error alone does not guarantee better performance. The consistency of predictions is also important. A model with a lower variance and standard deviation is generally more reliable.

Conclusion

- **Mean Error: Model A = 4, Model B = 4.625**
- **Variance: Model A = 1.5, Model B = 8.73**
- **Standard Deviation: Model A = 1.22, Model B = 2.95**
- Model A is more consistent because it has lower variance and standard deviation.